# Supplementary Material for *Timestep-Conditioned Adapter Scaling for Multi-view Diffusion Texture Generation*

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This supplement supports the revised 01549 manuscript. Fresh C is an image-space comparison on 300 UIDs disjoint from the original 01549 evaluation/probe IDs, the 1,118-object training list, and clean-v2 evaluation IDs. Its primary conditions and same-cohort addendum share one frozen adapter checkpoint, runner source, seed policy, inputs, target views, and metric implementation; the addendum is not an independent replication. Fresh C uses a controlled-retraining checkpoint and is not presented as a numerical reproduction of historical Table 4. Historical cohorts and interface checks are reported separately. Object is the inferential unit; multiple views and wrapper records do not increase $N$.

## S1. Figure asset credits and selection status

Main Figures 4–7 use Fresh C reference and method-level RGB outputs in a compact context-plus-shared-ROI layout. The original six-view, seven-condition panels remain in Supplementary Figure S1. The selected examples are outcome-ranked illustrations, not a representative sample. Image, checkpoint, run, seed, ROI, and output hashes are listed in `FRESH_C_FIGURE_SOURCE_ROI_MANIFEST.csv`.

The Fresh C No Adapter condition suppresses geometry-adapter correction. In the MVPainter wrapper, when `_adapter_scale` is unset the geometry residual is applied without a scale multiplier. These are different settings: No Adapter is not a rendered comparison against the unmodified native-default adapter path. Fresh C contains no separate output for that native-default path, so this part of the R1.1 request remains partially unresolved.

The current rights ledger records the selected Fresh C assets below as CC Attribution (CC BY 4.0), with figure use allowed when attributed. The complete credit line for each use is:

Table S1: Attribution for selected Fresh C assets in main Figures 5–7 and Supplementary Figure S1.

| Use | Full attribution |
|---|---|
| Figure 7(a), C3–GFL endpoint-favourable example | *Building 03* by Shalmon, via Sketchfab (`https://sketchfab.com/3d-models/c76ac44df995482185077da81939b306`), licensed CC BY 4.0 (`https://creativecommons.org/licenses/by/4.0/`). Fresh C target-view RGB outputs are cropped to the same ROI and arranged without color or detail adjustment. |
| Figure 5(a), C3-versus-fixed-scale example | *Octaèdre* by Segolene Delamare, via Sketchfab (`https://sketchfab.com/3d-models/0571e6db30c84b1593bc55a8e145a430`), licensed CC BY 4.0 (`https://creativecommons.org/licenses/by/4.0/`). Fresh C target-view RGB outputs are cropped to the same ROI and arranged without color or detail adjustment. |

| Use | Full attribution |
|---|---|
| Figure 5(b), Generic Linear counterexample | *Cube 01* by b0rF4x, via Sketchfab (`https://sketchfab.com/3d-models/06fa4974f8cd4d4ea792e98fa1521457`), licensed CC BY 4.0 (`https://creativecommons.org/licenses/by/4.0/`). Fresh C target-view RGB outputs are cropped to the same ROI and arranged without color or detail adjustment. |
| Figure 7(b) and Figure S1(a), adverse C3–GFL example | *Boat Low Poly Game Ready* by INinja, via Sketchfab (`https://sketchfab.com/3d-models/d0e116a62bfc4cdd9edf60671073b306`), licensed CC BY 4.0 (`https://creativecommons.org/licenses/by/4.0/`). Fresh C target-view RGB outputs are cropped to the same ROI and arranged without color or detail adjustment. |
| Figure 6, selected Fresh C color/detail illustration | *E5-5-2* by artsiyu, via Sketchfab (`https://sketchfab.com/3d-models/f6979d7ff8e54c4daf00fee46f09a139`), licensed CC BY 4.0 (`https://creativecommons.org/licenses/by/4.0/`). This previously viewed object came from a pre-existing C3–GFL FG-LPIPS stratum and was not selected for color. |

## S2. Complete Fresh C paired image-space statistics

Table S2 reports the complete seven-endpoint paired results for C3–GFL and each of the three named addendum contrasts. The primary C3–GFL FG-PSNR result is unadjusted as the single registered primary endpoint. The six registered secondary C3–GFL endpoints use their Holm family. The addendum's FG-PSNR and FG-LPIPS contrasts use the frozen six-test primary family; the other five endpoints per contrast use the separate 15-test secondary Core-7 family. All contrasts are mean A-minus-B differences with 10,000 object bootstrap resamples and pointwise two-sided percentile 95% CIs. Positive values favour A for PSNR/SSIM; negative values favour A for LPIPS. Adjusted $p$-values are from the frozen analysis authority. "A favourable" is the fraction of object-level differences in A's favourable direction and is descriptive.

Table S2: Complete paired object-level image-space results for the primary C3–GFL contrast and the three named same-cohort addendum contrasts. Differences are A minus B; favorable-object percentages respect endpoint direction.

| Endpoint | Mean difference [95% CI] | Median | A favorable | Adjusted $p$ |
|---|---|---|---|---|
| **C3 minus GFL** | | | | |
| FG-PSNR (dB) | +0.522 [+0.404, +0.641] | +0.555 | 68.7% | primary |
| FG-LPIPS | -0.005345 [-0.006326, -0.004403] | -0.004547 | 71.0% | $3.2 \times 10^{-22}$ |
| FG-SSIM | +0.02453 [+0.02120, +0.02805] | +0.02004 | 84.3% | $5.9 \times 10^{-33}$ |
| Edge-SSIM | +0.004494 [+0.003411, +0.005586] | +0.003613 | 69.0% | $9.4 \times 10^{-14}$ |
| Full-view PSNR (dB) | +0.271 [+0.170, +0.373] | +0.220 | 59.3% | $4 \times 10^{-07}$ |
| Full-view LPIPS | -0.003880 [-0.004926, -0.002873] | -0.003029 | 64.3% | $8 \times 10^{-13}$ |
| Full-view SSIM | +0.005619 [+0.004627, +0.006704] | +0.005205 | 83.0% | $7.9 \times 10^{-22}$ |
| **C3 minus Generic Linear** | | | | |
| FG-PSNR (dB) | -0.593 [-0.837, -0.351] | +0.074 | 51.7% | $7.2 \times 10^{-06}$ |
| FG-LPIPS | +0.013512 [+0.011601, +0.015487] | +0.012229 | 25.7% | $1.3 \times 10^{-32}$ |
| FG-SSIM | -0.02810 [-0.03197, -0.02439] | -0.02521 | 20.0% | $2.9 \times 10^{-35}$ |

| Endpoint | Mean difference [95% CI] | Median | A favorable | Adjusted $p$ |
|---|---|---|---|---|
| Edge-SSIM | -0.010693 [-0.013004, -0.008495] | -0.008967 | 33.7% | $1.6 \times 10^{-17}$ |
| Full-view PSNR (dB) | -0.503 [-0.686, -0.322] | -0.139 | 46.7% | $2.7 \times 10^{-07}$ |
| Full-view LPIPS | -0.000897 [-0.002578, +0.000862] | -0.001307 | 54.3% | 0.611 |
| Full-view SSIM | +0.000046 [-0.000984, +0.001011] | +0.001213 | 57.0% | 0.927 |
| **LLH minus C3** | | | | |
| FG-PSNR (dB) | +0.621 [+0.391, +0.855] | -0.153 | 46.7% | $1.2 \times 10^{-06}$ |
| FG-LPIPS | -0.012140 [-0.013861, -0.010480] | -0.010837 | 75.0% | $9 \times 10^{-34}$ |
| FG-SSIM | +0.01658 [+0.01311, +0.02003] | +0.01307 | 67.0% | $9 \times 10^{-18}$ |
| Edge-SSIM | +0.009033 [+0.006867, +0.011296] | +0.006741 | 63.3% | $2.3 \times 10^{-13}$ |
| Full-view PSNR (dB) | +0.816 [+0.639, +0.998] | +0.419 | 59.3% | $3.2 \times 10^{-16}$ |
| Full-view LPIPS | +0.004909 [+0.003266, +0.006510] | +0.004538 | 36.7% | $3.5 \times 10^{-08}$ |
| Full-view SSIM | -0.004330 [-0.005176, -0.003482] | -0.004563 | 27.7% | $5.9 \times 10^{-20}$ |
| **LLH minus Generic Linear** | | | | |
| FG-PSNR (dB) | +0.028 [-0.015, +0.070] | +0.017 | 52.3% | 0.201 |
| FG-LPIPS | +0.001372 [+0.000762, +0.002081] | +0.000176 | 46.7% | 0.000114 |
| FG-SSIM | -0.01152 [-0.01337, -0.00996] | -0.00945 | 8.0% | $2.6 \times 10^{-31}$ |
| Edge-SSIM | -0.001660 [-0.002096, -0.001256] | -0.001648 | 27.3% | $1.2 \times 10^{-12}$ |
| Full-view PSNR (dB) | +0.313 [+0.278, +0.347] | +0.304 | 87.3% | $2 \times 10^{-47}$ |
| Full-view LPIPS | +0.004011 [+0.003390, +0.004684] | +0.003904 | 18.0% | $1.7 \times 10^{-26}$ |
| Full-view SSIM | -0.004284 [-0.005217, -0.003543] | -0.003432 | 5.7% | $8.5 \times 10^{-20}$ |

## S3. Supporting layer/window and schedule evidence

A2 is exploratory discovery evidence; A3 is a higher-dose stress regime on the same underlying cohort. A3b is a bounded Fresh B layer/window analysis. Object-cluster tests detect response heterogeneity under the tested interventions, but they do not identify a dose-independent layer-by-time mechanism or an optimal schedule.

Table S3: Object-cluster Wald interaction summaries for the tested layer-by-window maps. Cell RMSE and interaction share describe these observed maps; they are not causal effect sizes.

| Map | Endpoint | Wald statistic | Interaction share; cell RMSE | Wald $p$ (A3b Holm) |
|---|---|---|---|---|
| A2 | FG-LPIPS | 481.081 | 43.5%; 0.001695 | $8.04 \times 10^{-99}$ |
| A2 | FG-PSNR | 1879.963 | 56.8%; 0.1222 dB | $< 10^{-300}$ |
| A3 | FG-LPIPS | 657.237 | 11.9%; 0.00643 | $1.14 \times 10^{-136}$ |
| A3 | FG-PSNR | 1389.118 | 0.8%; 0.207 dB | $1.28 \times 10^{-294}$ |
| A3b | FG-LPIPS | 225.015 | 14.0%; 0.000342 | $5.37 \times 10^{-43}$ |
| A3b | FG-PSNR | 525.700 | 36.0%; 0.0205 dB | $3.43 \times 10^{-107}$ |

The earlier A3b common-support calculation compared 10-step integrated raw $L_2$ norms across three layer groups. Because the saved correction tensors have different spatial resolutions (24×16, 48×32,

and 96×64) and the traces omit channel counts, absolute norms are dimension-dependent across groups. We withdraw the central-90% intervals and 0/2,250 overlap as cross-group dose-support evidence. The earlier development calibration also used cross-layer raw $L_2$ and is not interpreted as equal-dose targeting. The fixed A3b scale values remain the tested bounded profiles. A3b remains retrospective sensitivity evidence: its map describes response heterogeneity under the tested interventions, but does not identify a dose-independent layer-by-window effect.

The separate Fresh B cohort ($N = 150$) supplies schedule-location context for LLH, not evidence for C3. Table S4 preserves both registered LLH comparisons and the post-outcome HLL–LLL diagnostic. Paired intervals use 10,000 object-bootstrap resamples (seed 20261002). The LLH–LFM-EXACT intervals lie within the protocol's practical margins (FG-PSNR ±0.5 dB; FG-LPIPS ±0.01); this is a budget-explained interpretation within those margins, not exact equality. HLL–LLL is exploratory and its endpoint directions differ.

The Fresh B protocol-lock file states a freeze time of 2026-10-05 08:40 UTC, while the earliest preserved commit containing that file is dated 2026-10-06 00:52 UTC; its parent does not contain the file. This history does not establish when the lock was first created outside Git, but it cannot authenticate pre-run availability. Accordingly, Fresh B temporal contrasts, the A3b map, and the post-lock generic extension are treated as retrospective sensitivity evidence in this paper, even where their calculations follow the recorded comparison family. They are not described as verified preregistered confirmation.

Table S4: Fresh B paired object-level LLH comparisons, classified here as retrospective sensitivity because pre-run lock availability cannot be authenticated from the preserved history. Effects are A minus B; favorable rates respect endpoint direction. Holm values reproduce the recorded family.

| Contrast | Endpoint | Mean [95% CI] | A favors | $p$ / status |
|---|---|---|---|---|
| LLH–LFM-EXACT | FG-PSNR | +0.325 [+0.209,+0.446] dB | 54.0% | raw $\leq 0.0001$ |
| LLH–LFM-EXACT | FG-LPIPS | −0.00460 [−0.00563, −0.00357] | 76.0% | raw $\leq 0.0001$ |
| LLH–HLL | FG-PSNR | +0.687 [+0.404,+0.981] dB | 45.3% | $\leq 0.0004$; median −0.104 dB |
| LLH–HLL | FG-LPIPS | −0.01098 [−0.01349, −0.00851] | 80.7% | $\leq 0.0004$ |
| LLH–LLL | FG-PSNR | +0.734 [+0.597,+0.875] dB | 79.3% | $\leq 0.0004$ |
| LLH–LLL | FG-LPIPS | −0.00845 [−0.01029, −0.00659] | 82.0% | $\leq 0.0004$ |
| HLL–LLL (exploratory) | FG-PSNR | +0.047 [−0.120, +0.208] dB | 66.0% | 0.5696; median +0.476 dB |
| HLL–LLL (exploratory) | FG-LPIPS | +0.00253 [+0.00155,+0.00354] | 40.0% | $\leq 0.0002$; median +0.00139 |

The post-lock generic-schedule extension on this same Fresh B cohort is a retrospective sensitivity analysis, not independent confirmation. Table S5 reports the recorded budget-matched comparisons and endpoint-matched linear counter-result. Nominal requested-scale means do not equalize post-scale residual dose; these LLH comparisons do not establish dose-adjusted schedule superiority or strengthen C3 specifically.

Table S5: Post-lock Fresh B generic-schedule extension (same $N = 150$ cohort). Effects are LLH minus comparator. Budget-matched schedules form the locked eight-schedule family; endpoint-matched linear is the registered counter-comparison.

| Contrast | Endpoint | Mean [95% CI] | A favors | Holm $p$ |
|---|---|---|---|---|
| LLH–budget linear | FG-PSNR | +0.0967 [+0.0594,+0.1357] dB | 65.3% | Holm $p \leq 0.0008$ |
| LLH–budget linear | FG-LPIPS | -0.0009 [-0.0013,-0.0004] | 65.3% | Holm $p = 0.0008$ |
| LLH–budget cosine | FG-PSNR | +0.4629 [+0.3526,+0.5780] dB | 74.7% | Holm $p \leq 0.0008$ |
| LLH–budget cosine | FG-LPIPS | -0.0059 [-0.0071,-0.0046] | 80.0% | Holm $p \leq 0.0008$ |
| LLH–budget trapezoid | FG-PSNR | +0.4404 [+0.3334,+0.5521] dB | 74.0% | Holm $p \leq 0.0008$ |
| LLH–budget trapezoid | FG-LPIPS | -0.0057 [-0.0070,-0.0045] | 78.0% | Holm $p \leq 0.0008$ |
| LLH–budget Gaussian | FG-PSNR | +0.4711 [+0.3584,+0.5892] dB | 74.7% | Holm $p \leq 0.0008$ |
| LLH–budget Gaussian | FG-LPIPS | -0.0058 [-0.0071,-0.0045] | 79.3% | Holm $p \leq 0.0008$ |
| LLH–endpoint linear | FG-PSNR | -0.0373 [-0.0829,+0.0086] dB | – | Holm $p = 0.1150$ |
| LLH–endpoint linear | FG-LPIPS | +0.00109 [+0.0004,+0.0019] | – | Holm $p = 0.0024$ |

## S4. Requested/applied scale and residual-correction accounting

Table S6 lists representative Fresh C C3 records at five denoising steps. The native caps are 3.0 (deep), 3.5 (middle), and 0.8 (shallow). A cap-active fraction of 100% means all logged records in that group/step were clipped. Raw correction $L_2$ depends on both the residual and tensor dimensions, so it is not compared across layer groups. The table instead reports the median of wrapper-level mean absolute entries, a dimension-normalized descriptive summary; differences in feature scales and distributions still prevent interpreting it as an equal-dose intervention or quality outcome. The 750-row source accounting table contains all 50 steps, three groups, and five strategies.

Table S6: Representative Fresh C requested/applied scale and post-scale residual-correction accounting. The final column is the median wrapper-level mean absolute correction entry; it is dimension-normalized but remains descriptive. Wrappers, objects, and timesteps are not independent experimental units.

| Step | Layer group | Requested | Applied | Cap active | Median per-wrapper mean $\lvert c \rvert$ |
|---|---|---|---|---|---|
| 0 | Deep | 1.25 | 1.25 | 0% | 1.4786 |
| 0 | Middle | 1.25 | 1.25 | 0% | 0.3540 |
| 0 | Shallow | 1.25 | 0.80 | 100% | 0.0128 |
| 12 | Deep | 1.25 | 1.25 | 0% | 1.3969 |
| 12 | Middle | 1.25 | 1.25 | 0% | 0.4881 |
| 12 | Shallow | 1.25 | 0.80 | 100% | 0.0151 |
| 24 | Deep | 2.50 | 2.50 | 0% | 2.2588 |
| 24 | Middle | 2.50 | 2.50 | 0% | 1.1408 |
| 24 | Shallow | 2.50 | 0.80 | 100% | 0.0169 |
| 37 | Deep | 1.25 | 1.25 | 0% | 0.7513 |
| 37 | Middle | 1.25 | 1.25 | 0% | 0.4596 |
| 37 | Shallow | 1.25 | 0.80 | 100% | 0.0176 |
| 49 | Deep | 1.25 | 1.25 | 0% | 0.7354 |
| 49 | Middle | 1.25 | 1.25 | 0% | 0.1459 |
| 49 | Shallow | 1.25 | 0.80 | 100% | 0.0116 |

## S5. Historical and cross-interface boundaries

The completed `MAIN-STAGE-276` follow-up contains 1,104 records (276 objects by four schedules), with 50 Euler steps, seed 42, unique-six target views, shared initial latents, and a common runner within the run. Its final run manifest records completion with exit code 0. The follow-up was specified after inspection of earlier clean-v2 results and was not blind. It is reported separately from Fresh C and is not an independent replication of the original 01549 development probe.

Here, temporal L–L–H means that the requested values are low in early and middle steps and high in late steps. This is distinct from the layer-conditioned Fresh C LLH addendum condition described in the main paper. With 50 steps, the code assigns 17/16/17 steps to early/middle/late. C3 therefore has deep/middle applied-scale sums of 82.5; temporal L–L–H has 83.75 because its high stage occupies 17 rather than 16 steps; the shallow applied sum is 40 for both after the 0.8 cap. The fixed-mean comparator has a deep/middle sum of 83.33 and a shallow sum of 40. These are near-matched scalar sums, not exactly equal sums or matched residual-norm doses.

Table S7 gives all six formal paired endpoints for C3 versus temporal L–L–H and C3 versus the fixed-mean comparator. Effects are C3 minus comparator; positive values favor C3 for PSNR/SSIM, and negative values favor C3 for LPIPS. Temporal L–L–H outperforms C3 on all six formal endpoints in this post-inspection stage-placement experiment: it has higher PSNR/SSIM and lower LPIPS endpoints. In contrast, C3 exceeds fixed-mean on five of six formal endpoints. These results show that the measured response varies with tested stage placement; they do not establish a universal schedule winner. The contrasts do not isolate placement from the small applied-scale-sum differences or the resulting residual-state trajectory, and post-scale residual norms are not matched.

Table S7: Formal paired comparisons from the strict-276 stage-placement follow-up. Effects are C3 minus comparator; 95% percentile-bootstrap intervals are object-level.

| Contrast | Endpoint | Mean difference [95% CI] |
|---|---|---|
| **C3 minus temporal L–L–H** | | |
| | Full-view PSNR (dB) | $-0.268100$ $[-0.311101, -0.227797]$ |
| | FG-PSNR (dB) | $-0.177084$ $[-0.230539, -0.130784]$ |
| | FG-SSIM | $-0.008851$ $[-0.011455, -0.006291]$ |
| | Edge-SSIM | $-0.035301$ $[-0.036581, -0.034031]$ |
| | FG-LPIPS | $+0.001896$ $[+0.001205, +0.002578]$ |
| | Full-view LPIPS | $+0.008658$ $[+0.007465, +0.009831]$ |
| **C3 minus fixed-mean** | | |
| | Full-view PSNR (dB) | $+0.556369$ $[+0.525520, +0.589220]$ |
| | FG-PSNR (dB) | $+0.502875$ $[+0.462018, +0.542244]$ |
| | FG-SSIM | $+0.032083$ $[+0.028966, +0.035172]$ |
| | Edge-SSIM | $+0.001696$ $[+0.000611, +0.002782]$ |
| | FG-LPIPS | $-0.003797$ $[-0.004341, -0.003260]$ |
| | Full-view LPIPS | $+0.001420$ $[+0.000778, +0.002072]$ |

The saved-artifact Full-SSIM branch is kept separate from the formal pre-save metrics above: C3 minus temporal L–L–H is $-0.001697$ (95% CI $[-0.002047, -0.001340]$); C3 minus fixed-mean is $+0.007486$ ($[+0.006736, +0.008262]$). These saved-artifact values are not pooled with the formal metric column.

A retrospective strict-276 Core-7 summary uses the clean-v2 object list (SHA-256 `a6aa8ab6e475763e1b5e67dc8c712ec8f1940e3952d65897c820887554bbd044`) and the same recorded checkpoint SHA as Fresh C. It shares the 276-object cohort/checkpoint lineage with the separate `MAIN-STAGE-276` profile-placement follow-up, so those results are not independent cohort replications. The row audit verifies 276 unique object indices per condition and identical object labels by index. However, Core-7 does not establish one common runner or fully shared inputs across all seven conditions: No Adapter and GFH are the two newly rerun arms with per-object input-tensor hashes; their recorded runner hash is `ae5d5664e7d0ebcd4bd84c7c7d688c4a8962c6d74463fe88e7a866404911e7e0`. C3 records a different runner hash, `d4af0c1d7f42168a585b505a6d775ea3ff7ffb18345156ebf358f272025fa7f2`; GFL and the three layer conditions do not establish the same runner hash, and their serialized rows do not contain per-object input-tensor hashes. Thus Core-7 is a retrospective diagnostic, not a same-runner confirmatory comparison. It is not pooled with Fresh C, and the shared checkpoint hash does not make it a numerical reproduction of the original 01549 Table 4 result.

Table S8 gives the seven-condition means for the requested image-space endpoints. These stored metrics show an endpoint trade-off: C3 has a higher Full-view PSNR mean than GFH but a lower FG-PSNR mean. The Core-7 Layer LLH row is distinct from Fresh C's layer-conditioned LLH addendum and Fresh B's temporal LLH condition. Table S9 reports selected object-indexed contrasts. C3–GFH intervals were recalculated from the serialized per-object rows after inspection of the Core-7 summary; they are post-hoc, nominal pointwise intervals. The other listed contrasts reproduce the separate Core-7 paired-statistics file. The intervals describe the saved metric rows and do not overcome the common-runner/input-hash limitations.

Table S8: Retrospective Core-7 strict-276 condition means. Rows contain 276 object-level metrics per condition. The collection has incomplete common runner/input identity as described above; means are descriptive and are not Fresh C or an independently verified same-runner comparison.

| Condition | Full-view PSNR↑ | FG-PSNR↑ | FG-LPIPS↓ | FG-SSIM↑ | Edge-SSIM↑ |
|---|---|---|---|---|---|
| No Adapter | 10.563 | 8.826 | 0.2059 | 0.4799 | 0.4794 |
| GFL | 18.229 | 10.731 | 0.1917 | 0.3987 | 0.5201 |
| GFH | 18.242 | 13.153 | 0.1677 | 0.4783 | 0.5538 |
| C3 | 18.968 | 11.939 | 0.1817 | 0.4296 | 0.5278 |
| Layer Fixed Mean | 19.516 | 13.100 | 0.1698 | 0.4550 | 0.5363 |
| Layer LHL | 19.635 | 12.974 | 0.1719 | 0.4445 | 0.5273 |
| Layer LLH | 20.828 | 13.975 | 0.1610 | 0.4620 | 0.5476 |

Table S9: Selected retrospective Core-7 paired contrasts (first condition minus second). Intervals are pointwise 95% object-bootstrap percentile intervals with 10,000 resamples and seed 20260930; they are unadjusted and descriptive. The C3–GFH intervals were recalculated post hoc. The final column counts objects where the first condition is favorable for PSNR.

| Contrast | Endpoint | Mean difference [95% CI] (dB) | Objects favoring first |
|---|---|---|---|
| C3–GFH | Full-view PSNR | $+0.725229$ $[+0.647134, +0.806467]$ | 240/276 |
| C3–GFH | FG-PSNR | $-1.214235$ $[-1.313952, -1.114705]$ | 25/276 |
| C3–GFL | FG-PSNR | $+1.207771$ $[+1.116967, +1.294989]$ | 259/276 |
| LLH–GFH | FG-PSNR | $+0.822292$ $[+0.616795, +1.033035]$ | 148/276 |
| LLH–C3 | FG-PSNR | $+2.036527$ $[+1.806804, +2.260756]$ | 227/276 |

The frozen MV-Adapter layer/window probe analyzed 98 of 99 planned objects; the omitted input was a GLB with path geometry and no triangle faces, so no substitute was used. Its production additive-residual bootstrap had an invalid null and is not used for interaction inference. The post hoc object-cluster robust Wald reanalysis found no interaction surviving its exploratory four-metric Holm correction (FG-LPIPS Holm $p = 0.303$; PSNR $p = 0.868$; FG-SSIM $p = 0.156$; CIEDE2000 $p = 0.862$). This does not establish absence of interaction or equivalence.

A separate 76-object Exact-holdout MV-Adapter panel adds layer-wise fixed, L–HL, and L–LH outputs to existing global controls. It scales condition-guider features before UNet down-block residual injection and does not reproduce the GeoTex ResNet residual interface. The wrapper matched the existing global runner bitwise in a 3-object identity audit, but this is not an identity comparison against the unmodified official pipeline at native scale. Table S10 gives the seven condition means and Table S11 gives all five paired contrasts across six endpoints. Layer-wise/global mean differences are metric-dependent; object medians and favorable fractions also vary. Within-profile temporal comparisons show one pointwise interval excluding zero in each comparison, with the other five including zero. These 30 intervals are unadjusted and descriptive. The protocol and result files first appear in one preserved commit, so we classify this analysis as retrospective sensitivity rather than confirmation. It establishes neither equivalence nor transfer of TCAS.

The MVDiffusion schedule panel contains 525 object-condition rows: seven conditions over the same 75 objects. All six scheduled-condition per-object CSVs match their rows in the combined panel. The runs generated 12-view output grids; the frozen metric manifest scores six target views per object. The archived paired-bootstrap output used positive values to favor the left condition; for lower-is-better FG-LPIPS, CIEDE2000, and GT-relative texture error, those values are improvement-coded ($B - A$), not raw $A - B$. We independently recomputed the five planned contrasts from the 75 paired object rows and now report raw $A - B$ in Table S12. For L–LLH minus G–FL, CIEDE2000 is $+1.373021$ (95% CI $[+0.418244, +2.374678]$), so the layer-wise setting has higher color error; PSNR, FG-SSIM, and Edge-SSIM also favor G–FL, GT-relative texture error is lower for L–LLH, and FG-LPIPS does not separate. The corresponding LLH/LHL

layer-wise-versus-global contrasts have the same endpoint direction pattern. Within the global LLH/LHL comparison, the single pointwise FG-LPIPS interval favors G–LLH; the other five endpoints include zero. All 30 intervals are pointwise and unadjusted. GT-relative texture error summarizes agreement with three reference variation statistics and is not a fidelity measure. A correction note and the full recomputation outputs accompany the preserved original analysis files in the author-review package.

MVDiffusion scales correspondence-aware CPBlock outputs, while depth is concatenated into the latent and left unscaled. The 75-object schedule panel used a compatibility deployment base. A later check found 1,552/1,552 loaded tensors and 12/12 rendered views for one object identical to a native SD2-depth mirror; the full panel was not rerun. These results remain separate, mixed endpoint evidence on a different conditioning path; no cross-interface score pooling, architecture-causal claim, or general cross-backbone result is made.

The first-round 01549 Section 4.7 adapter-dependence analysis is retained as a separate within-family result. It reported three outcomes for uniform high scale across adapter regimes: texture flattening without per-layer caps, high-frequency artifact amplification under a strong-residual adapter with per-layer caps, and negligible effect for a weakened adapter. Its same-checkpoint cap ablation reported both flattening and amplification, and its residual-magnitude sweep ($0.1\times$–$3\times$ output projection) reported increasing intervention strength with residual magnitude. A six-object stage ablation reported that the most sensitive stage differed between the main and artifact-prone adapters. These first-round results were explicitly described as adapter-dependent; they are retained as historical, within-architecture evidence and do not answer the current cross-backbone question.

A separate 76-object MV-Adapter study tested layer-wise L–LLH under an alternative budget-neutral contiguous partition B′ (per-point multipliers $[0.52018, 0.52018, 1.47982, 1.47982]$, weighted mean 1). Relative to the corresponding global controls, the B′ results are mixed across endpoints: the layer condition favors the global controls on FG-SSIM, favors the layer condition on the benefit-oriented CIEDE2000 and GT-relative texture summaries, and does not separate on FG-PSNR, Edge-SSIM, or FG-LPIPS. Its difference from the original partition is not resolved on any reported metric. These data describe this MV-Adapter layer intervention only and are not evidence for the TCAS timestep profile or for a general mapping rule.

The companion M3 profile uses $[0.41953, 0.41953, 1.19349, 1.19349]$, whose mean multiplier is approximately 0.8065. It changes both group assignment and total nominal residual budget, so it is not a pure mapping comparison with B′. Its audit classified 16 of 18 direction summaries as stable relative to the original partition, with one small Edge-SSIM attenuation and one near-zero FG-LPIPS sign change. Because the preserved protocol and result rows first appear in the same commit, M3 is retained as retrospective sensitivity. Neither M3 nor B′ establishes mapping robustness or cross-backbone generalization.

**Retrospective MV-Adapter layer/time panel.** A separate Exact-holdout analysis combines existing global controls with three new layer-wise conditions (L-FIX, L-LHL, L-LLH), totaling 228 new outputs on 76 objects. The wrapper scales four condition-guider feature tensors before they enter the UNet as down-block residuals; this is not the GeoTex ResNet residual-correction interface. Its 3-object identity audit matched the wrapper to the existing global runner on 9/9 schedule/object cases, but does not test the unmodified official pipeline at native scale 1.0. Layer-wise versus global contrasts show metric-dependent directions: PSNR, CIEDE2000, and ground-truth-relative texture error favor layer-wise rows, while FG-/Edge-SSIM favor global rows; FG-LPIPS is not separated. Within-profile temporal comparisons have no consistent endpoint pattern. The 10,000-resample intervals below are pointwise and descriptive, without family-wise correction across 30 metric/comparison cells. The earliest preserved commit adds both protocol and outcomes, so the panel is treated as retrospective rather than confirmatory. It does not establish equivalence or TCAS transfer.

Table S10: MV-Adapter Exact-holdout condition means for the 76-object seven-condition panel. These are descriptive interface-specific image metrics and are not pooled with GeoTex or MVDiffusion.

| Condition | PSNR↑ | FG-SSIM↑ | Edge-SSIM↑ | FG-LPIPS↓ | CIEDE2000↓ | GT-texture error↓ |
|---|---|---|---|---|---|---|
| R0 | 13.114180 | 0.498481 | 0.233429 | 0.183496 | 20.719805 | 1.700798 |
| G-FL | 13.357177 | 0.527539 | 0.239997 | 0.187693 | 20.149543 | 2.224047 |
| G-LHL | 13.344064 | 0.528325 | 0.240235 | 0.188051 | 20.182146 | 2.269096 |
| G-LLH | 13.353306 | 0.527814 | 0.239974 | 0.188346 | 20.157122 | 2.198930 |
| L-FIX | 13.431922 | 0.523473 | 0.239052 | 0.187647 | 19.877910 | 1.903374 |
| L-LHL | 13.446348 | 0.525148 | 0.239619 | 0.187920 | 19.868912 | 1.980378 |
| L-LLH | 13.453797 | 0.523564 | 0.238899 | 0.187731 | 19.805120 | 1.913272 |

Table S11: All five reported MV-Adapter paired contrasts. Deltas are direction-adjusted so positive favors the left condition (lower-is-better metrics are sign-reversed). Intervals are pointwise object-bootstrap percentiles (10,000 resamples; seed 20260928), descriptive and unadjusted.

| Contrast | Endpoint | Mean Δ [95% CI] | Median Δ | Left favorable |
|---|---|---|---|---|
| P1: L-LLH vs. G-FL | PSNR↑ | +0.096619 [+0.014894, +0.186322] | +0.038948 | 60.5% |
| P1: L-LLH vs. G-FL | FG-SSIM↑ | −0.003975 [−0.007222, −0.001110] | −0.000055 | 47.4% |
| P1: L-LLH vs. G-FL | Edge-SSIM↑ | −0.001098 [−0.002246, −0.000084] | −0.000068 | 48.7% |
| P1: L-LLH vs. G-FL | FG-LPIPS↓ | −0.000038 [−0.002249, +0.001836] | −0.000145 | 43.4% |
| P1: L-LLH vs. G-FL | CIEDE2000↓ | +0.344422 [+0.162359, +0.548366] | +0.074472 | 67.1% |
| P1: L-LLH vs. G-FL | GT-texture error↓ | +0.310776 [+0.128347, +0.523530] | −0.008452 | 44.7% |
| P2: L-LLH vs. G-LLH | PSNR↑ | +0.100491 [+0.016411, +0.193981] | +0.033523 | 60.5% |
| P2: L-LLH vs. G-LLH | FG-SSIM↑ | −0.004249 [−0.007467, −0.001399] | −0.000938 | 46.1% |
| P2: L-LLH vs. G-LLH | Edge-SSIM↑ | −0.001075 [−0.002200, −0.000087] | −0.000020 | 47.4% |
| P2: L-LLH vs. G-LLH | FG-LPIPS↓ | +0.000615 [−0.001370, +0.002409] | +0.000046 | 50.0% |
| P2: L-LLH vs. G-LLH | CIEDE2000↓ | +0.352002 [+0.161829, +0.565100] | +0.083198 | 63.2% |
| P2: L-LLH vs. G-LLH | GT-texture error↓ | +0.285659 [+0.092803, +0.508436] | −0.029244 | 46.1% |
| P3: L-LLH vs. L-FIX | PSNR↑ | +0.021875 [−0.021181, +0.064134] | −0.003147 | 43.4% |
| P3: L-LLH vs. L-FIX | FG-SSIM↑ | +0.000092 [−0.001301, +0.001429] | −0.000029 | 46.1% |
| P3: L-LLH vs. L-FIX | Edge-SSIM↑ | −0.000152 [−0.001051, +0.000628] | −0.000030 | 47.4% |
| P3: L-LLH vs. L-FIX | FG-LPIPS↓ | −0.000084 [−0.000821, +0.000518] | +0.000126 | 59.2% |
| P3: L-LLH vs. L-FIX | CIEDE2000↓ | +0.072790 [+0.004371, +0.156696] | −0.004923 | 48.7% |
| P3: L-LLH vs. L-FIX | GT-texture error↓ | −0.009898 [−0.088568, +0.077841] | +0.002578 | 52.6% |
| P4: L-LHL vs. G-LHL | PSNR↑ | +0.102285 [+0.022567, +0.195757] | +0.048007 | 67.1% |
| P4: L-LHL vs. G-LHL | FG-SSIM↑ | −0.003177 [−0.005872, −0.000730] | −0.000283 | 47.4% |
| P4: L-LHL vs. G-LHL | Edge-SSIM↑ | −0.000615 [−0.001482, +0.000167] | +0.000066 | 52.6% |
| P4: L-LHL vs. G-LHL | FG-LPIPS↓ | +0.000131 [−0.002170, +0.002094] | −0.000162 | 44.7% |
| P4: L-LHL vs. G-LHL | CIEDE2000↓ | +0.313234 [+0.153138, +0.486864] | +0.094149 | 67.1% |
| P4: L-LHL vs. G-LHL | GT-texture error↓ | +0.288718 [+0.104317, +0.504549] | −0.010427 | 48.7% |
| P5: G-LLH vs. G-LHL | PSNR↑ | +0.009242 [−0.022729, +0.055429] | +0.000649 | 53.9% |

| Contrast | Endpoint | Mean Δ [95% CI] | Median Δ | Left favorable |
|---|---|---|---|---|
| P5: G-LLH vs. G-LHL | FG-SSIM↑ | −0.000511 [−0.001207, +0.000102] | +0.000114 | 60.5% |
| P5: G-LLH vs. G-LHL | Edge-SSIM↑ | −0.000261 [−0.000522, −0.000032] | +0.000029 | 53.9% |
| P5: G-LLH vs. G-LHL | FG-LPIPS↓ | −0.000296 [−0.001198, +0.000712] | −0.000079 | 36.8% |
| P5: G-LLH vs. G-LHL | CIEDE2000↓ | +0.025024 [−0.032059, +0.108336] | −0.003118 | 42.1% |
| P5: G-LLH vs. G-LHL | GT-texture error↓ | +0.070166 [−0.050221, +0.216403] | −0.003606 | 43.4% |

Table S12: MVDiffusion 75-object paired contrasts, reported as raw condition A minus condition B. Every comparison has $N = 75$ paired objects; intervals are pointwise 95% object-bootstrap percentile CIs (10,000 resamples, seed 20260928), without multiplicity adjustment. Higher PSNR, FG-SSIM, and Edge-SSIM are favorable; lower FG-LPIPS, CIEDE2000, and GT-relative texture error are favorable. The last metric is a mean symmetric-log error across three reference-relative variation statistics and is not a texture-fidelity measure.

| Contrast (A − B) | Endpoint | Raw mean difference [95% CI] |
|---|---|---|
| **P1: L-LLH − G-FL** | | |
| | PSNR (dB) | −0.453828 [−0.752102, −0.160451] |
| | FG-SSIM | −0.019393 [−0.029361, −0.009856] |
| | Edge-SSIM | −0.005941 [−0.010305, −0.001382] |
| | FG-LPIPS | +0.002745 [−0.001808, +0.007632] |
| | CIEDE2000 | +1.373021 [+0.418244, +2.374678] |
| | GT-relative texture error | −0.082286 [−0.145316, −0.020073] |
| **P2: L-LLH − G-LLH** | | |
| | PSNR (dB) | −0.466728 [−0.767571, −0.174846] |
| | FG-SSIM | −0.019245 [−0.029244, −0.009725] |
| | Edge-SSIM | −0.006207 [−0.010601, −0.001620] |
| | FG-LPIPS | +0.002998 [−0.001458, +0.007765] |
| | CIEDE2000 | +1.377180 [+0.400867, +2.393609] |
| | GT-relative texture error | −0.064768 [−0.124369, −0.005093] |
| **P3: L-LLH − L-FIX** | | |
| | PSNR (dB) | −0.092564 [−0.182342, −0.003231] |
| | FG-SSIM | +0.000528 [−0.003700, +0.004874] |
| | Edge-SSIM | −0.002828 [−0.005185, −0.000405] |
| | FG-LPIPS | −0.001089 [−0.002704, +0.000499] |
| | CIEDE2000 | +0.082626 [−0.235252, +0.410153] |
| | GT-relative texture error | −0.037090 [−0.081113, +0.000258] |
| **P4: L-LHL − G-LHL** | | |
| | PSNR (dB) | −0.520646 [−0.810872, −0.232104] |
| | FG-SSIM | −0.021431 [−0.032306, −0.011114] |
| | Edge-SSIM | −0.007282 [−0.011359, −0.003234] |
| | FG-LPIPS | +0.002491 [−0.001895, +0.007174] |
| | CIEDE2000 | +1.545280 [+0.592682, +2.526751] |
| | GT-relative texture error | −0.098921 [−0.154601, −0.044345] |
| **P5: G-LLH − G-LHL** | | |
| | PSNR (dB) | −0.078763 [−0.163889, +0.004507] |
| | FG-SSIM | −0.0000003 [−0.002323, +0.002393] |
| | Edge-SSIM | −0.001471 [−0.003510, +0.000589] |
| | FG-LPIPS | −0.001371 [−0.002642, −0.000141] |
| | CIEDE2000 | +0.015861 [−0.264829, +0.299044] |
| | GT-relative texture error | −0.032667 [−0.068249, +0.002649] |

## S6. Historical N=20 unseen-view results

Table S13 reports the bounded historical N=20 GLB results. The subset comprises 20 UV-compatible objects from the original 24-object pool; four inputs were excluded for lack of UVs. Each object is the paired unit after averaging 11 frozen unseen views. Rendering uses unlit base color with native sampler interpretation. Endpoint directions are mixed; these statistics do not establish population-level 3D superiority, PBR/material fidelity, seam quality, or human preference. The source GLB assets and internal montage are not included in this supplement.

Table S13: Historical N=20 unlit base-color paired unseen-view summaries (C3 minus comparator). The 20-object UV-compatible subset comes from the historical 24-object pool; four objects were excluded for missing UVs. Each object is averaged over 11 frozen unseen views. Intervals are 10,000-object-bootstrap percentile CIs (seed 20261005), descriptive and unadjusted.

| Comparator | Endpoint | Mean difference [95% CI] | Objects favoring C3 |
|---|---|---|---|
| GFL | FG-PSNR (dB) | -0.066 [-0.413, +0.290] | 5/20 (25%) |
| GFL | FG-LPIPS | -0.004719 [-0.007504, -0.002544] | 17/20 (85%) |
| GFL | CIEDE2000 | +0.133 [-1.161, +1.319] | 6/20 (30%) |
| GFH | FG-PSNR (dB) | +0.084 [-0.290, +0.426] | 14/20 (70%) |
| GFH | FG-LPIPS | +0.007709 [+0.004230, +0.012101] | 4/20 (20%) |
| GFH | CIEDE2000 | -1.027 [-2.224, +0.229] | 14/20 (70%) |
| No Adapter | FG-PSNR (dB) | +3.135 [+1.721, +4.493] | 18/20 (90%) |
| No Adapter | FG-LPIPS | -0.014519 [-0.029860, -0.001326] | 16/20 (80%) |
| No Adapter | CIEDE2000 | -12.508 [-18.556, -6.529] | 18/20 (90%) |

## S7. Selected negative visual evidence and unresolved 3D checks

Figure S1 presents one selected illustrative Fresh C case where GFL is adverse to C3 on the paired image-space comparison. It does not estimate failure prevalence or color-correction performance. Full attribution appears in Table S1.




Figure S1: Selected illustrative adverse C3–GFL case from the rights-cleared Fresh C 2D pool. It is not a representative estimate and does not support color-correction, PBR, seam, or human-preference claims.

The attempted Fresh C 24-object GLB bake did not pass its input/output gate: 9/24 sources had out-of-range native UVs, and 5 assets / 27 primitives used unsupported BLEND. No valid Fresh C 24-object bake results were generated. The seam endpoint is retired because the required seam truth set and native full-panel validation did not pass. No full PBR comparison is available. These checks are limitations, not evidence of method failure or success.

## S8. Statistical authority and reproduction map

Each numerical result in this Supplement is mapped to the internal claim ledger and scoped result sources. Fresh C primary contrasts use object-level deltas and a locked analysis; requested/applied-scale records use the per-step accounting table. Layer/window and Fresh B results are maintained in separate source summaries, with Fresh B classified as retrospective sensitivity because its lock chronology cannot be authenticated from preserved Git history. The texture-complexity cohort and the color-diagnostic set are separate from Fresh C and from each other. The historical N=20 GLB-native summary has a separate analysis record; no N=20 image assets are redistributed. Original Table 9 and Table 3 below are reproduced only as historical source text with explicit provenance limits; neither is treated as a newly verified current result.

**Fresh C identity details.** The 300-object Fresh C list is disjoint by UID from the original eval300, including the 24-object development probe, the 1,118-object training list, and the clean-v2 eval300. The primary four-condition run and same-cohort Generic Linear/LLH addendum use the same frozen checkpoint SHA-256 `0618d6b284ab47aa16d7a89dc447f5ba4455ff6c9d918e5e129478a99e0114c0`, runner source SHA-256 `e5c28e915ba137d3541eb4c032daaf62a798332d63b32917b749fe1bcc74e9e3`, cohort, input tensors, endpoint implementation, and seed policy. The seed is $42 +$ `object_idx` and is shared across the matched conditions for each object. Primary and addendum integrity gates pass for 1,200/1,200 and 600/600 condition rows. Fresh C uses the recorded deep/middle/shallow caps of 3.0/3.5/0.8, with requested, applied, and residual scales logged separately. This is a held-out paired comparison under the stated protocol. The historical identity remains explicit: 01549 names v1 weights for its main tables and a separate v2 strong-residual, per-layer-capped instance for Section 4.7/FAC. Fresh C uses a controlled retraining and is not presented as a numerical reproduction of the old Table 4 +0.96 dB estimate.

## S9. Historical 01549 Table 9: strict-276 schedule transfer

The 01549 submission reports this schedule-transfer comparison on the 276 objects disjoint from the 24-object probe. It uses the strong-residual, per-layer-capped v2 adapter, not the v1 checkpoint named for 01549 Table 4. The values below are reproduced from the authenticated 01549 submission; they are not pooled with Fresh C or the separate MAIN-STAGE-276 profile-placement analysis. The table therefore documents schedule transfer on its stated adapter instance and remains distinct from Fresh C's matched single-checkpoint comparison.

Table S14: Values reproduced from original 01549 Table 9 for the stated strict-276 retrospective schedule-transfer comparison. The adapter/run differs from Fresh C; row-level runner and input provenance are incomplete. Values are preserved as submitted historical results, not independently verified statistics.

| Schedule | FG-SSIM | PSNR | C3–schedule ΔPSNR (95% CI) | C3-favourable objects |
|---|---|---|---|---|
| Trapezoid | 0.356 | 19.04 | +0.258 [+0.207, +0.308] | 206/276 |
| Gaussian peak | 0.358 | 18.97 | +0.324 [+0.281, +0.367] | 232/276 |
| Linear warm-up | 0.377 | 18.97 | +0.323 [+0.263, +0.386] | 214/276 |
| Cosine bump | 0.330 | 18.39 | +0.902 [+0.824, +0.981] | 253/276 |
| C3 (TCAS) | 0.376 | 19.29 | – | – |

## S10. Historical 01549 Table 3: extended 24-object development diagnostics

The table below is transcribed from the authenticated 01549 source and moved from the main text to preserve the original 24-object development record. The candidate-level row outputs and selection-time provenance are unavailable for independent reanalysis. Laplacian variance, RGB standard deviation, gradient magnitude, and high-frequency energy are variation diagnostics; this historical table does not establish texture fidelity, a population effect, or an optimal schedule.

Table S15: Values transcribed from original 01549 Table 3 for representative variants in the 24-object development probe. These are historical values, not independently recomputed statistics; variation measures do not establish texture fidelity.

| Variant | ΔLap Var | ΔRGB Std | ΔGrad | ΔHF Energy |
|---|---|---|---|---|
| C3 | −0.001 | −0.023 | +0.142 | +0.058 |
| C4 | −0.012 | −0.032 | +0.086 | +0.062 |
| A_s2.50 | −0.011 | −0.044 | +0.054 | +0.078 |
| A_s2.00 | −0.006 | −0.040 | +0.100 | +0.070 |
| A_s1.25 | −0.005 | −0.032 | +0.104 | +0.053 |

## S11. First-round FAC controlled re-examination

The 01549 first-round package reports seven trained FAC configurations under the paired 300-object protocol. All values below are retained as originally reported; this is not a new 1008 training run. The 12-object untrained C3-envelope control is separate and is described in the main text.

Table S16: FAC dose-response table reported in the first-round 01549 supplement. FG-PSNR and FG-SSIM are the reported absolute values; ΔFG-PSNR and wins compare each trained configuration with TCAS.

| Configuration | Temporal gate | FG-PSNR | FG-SSIM | ΔFG-PSNR | Wins |
|---|---|---|---|---|---|
| TCAS baseline (C3) | fixed C3 | 9.88 | 0.3681 | – | – |
| LTAG only | joint, unconstrained | 7.79 | 0.3487 | −2.08 | 4/300 |
| LTAG+GSG | joint, unconstrained | 7.03 | 0.3163 | −2.84 | 6/300 |
| Full FAC | joint, unconstrained | 6.94 | 0.3204 | −2.94 | 5/300 |
| Two-stage + GSG/FSC | frozen at C3 | 7.32 | 0.3074 | −2.56 | 4/300 |
| Envelope penalty ($\lambda = 1$) | joint, penalized to C3 | 7.14 | 0.3021 | −2.74 | 2/300 |
| Fidelity weighting | frozen at C3 | 7.51 | 0.3110 | −2.37 | 5/300 |
| Gentle schedule | frozen at C3 | 8.77 | 0.4175 | −1.10 | 48/300 |

The submitted protocol states that only the LTAG/GSG/FSC controllers (roughly $6 \times 10^3$ parameters) were trained; the base model, VAE, and GeoTex-Adapter remained frozen. It specifies the enhanced latent denoising objective and 2,000 AdamW steps on a training pool stated as 1,706 objects. The separate 12-object untrained control was reported at ΔFG-PSNR = −0.18 dB (95% CI $[-0.42, +0.05]$). A recovered configuration instead lists 1,200 training objects, and no run-bound seed/output manifest was found. We

retain the original values as reported, disclose this reproducibility limit, and make no code-release claim.

## S12. Retrospective texture-complexity boundary

This analysis uses the separate FRESH_CONFIRM_300 cohort ($N = 300$), not Fresh C or Fresh B. LLH and native GFL rows match the frozen object list and share the recorded checkpoint, runner, seed, six unique target views, sampler, and protocol. The paired LLH–GFL mean differences are +1.2877 dB FG-PSNR (95% object-bootstrap CI $[+0.9383, +1.6346]$; 57.7% LLH-favorable objects) and −0.01650 FG-LPIPS ($[-0.01887, -0.01413]$; 73.3% LLH-favorable objects).

The descriptor analysis tests two image-space outcomes against five ground-truth descriptors: foreground Laplacian variance, RGB standard deviation, gradient magnitude, high-frequency energy, and coverage. Spearman correlations use average ranks and paired-object percentile bootstrap intervals (10,000 resamples, seed 20261002). The family contains ten plus-one-corrected bootstrap sign tests and Holm correction across all ten. Nine tests reject at 0.05; the only non-rejection is FG-LPIPS versus coverage (adjusted $p = 0.0508$). The preserved H5 hypothesis excerpt predates one archived outcome bundle, but the main multiplicity lock classifies this family as retrospective after prior outcome exposure; that chronology has not been reconciled. We therefore report all associations as retrospective sensitivity evidence, not as preregistered confirmation.

Table S17: Retrospective Spearman associations between LLH–GFL image-space deltas and ground-truth descriptors in FRESH_CONFIRM_300. Correlation intervals are paired-object bootstrap intervals; adjusted $p$ values use the ten-test Holm family with plus-one-corrected bootstrap sign tests.

| Outcome delta | Ground-truth descriptor | $\rho$ [95% CI] | Holm $p$ | Decision |
|---|---|---|---|---|
| FG-PSNR | Laplacian variance | $-0.524$ $[-0.603, -0.435]$ | 0.0020 | reject |
| FG-PSNR | RGB standard deviation | $-0.380$ $[-0.488, -0.266]$ | 0.0020 | reject |
| FG-PSNR | Gradient magnitude | $-0.464$ $[-0.548, -0.371]$ | 0.0020 | reject |
| FG-PSNR | High-frequency energy | $-0.691$ $[-0.745, -0.626]$ | 0.0020 | reject |
| FG-PSNR | Foreground coverage | $-0.134$ $[-0.241, -0.026]$ | 0.0352 | reject |
| FG-LPIPS | Laplacian variance | $+0.604$ $[+0.511, +0.681]$ | 0.0020 | reject |
| FG-LPIPS | RGB standard deviation | $+0.379$ $[+0.272, +0.482]$ | 0.0020 | reject |
| FG-LPIPS | Gradient magnitude | $+0.563$ $[+0.469, +0.643]$ | 0.0020 | reject |
| FG-LPIPS | High-frequency energy | $+0.677$ $[+0.599, +0.744]$ | 0.0020 | reject |
| FG-LPIPS | Foreground coverage | $-0.118$ $[-0.233, +0.0002]$ | 0.0508 | not rejected |

Ground-truth foreground Laplacian-variance quartiles contain 75 objects each. The FG-PSNR mean LLH–GFL difference declines from +4.140 dB in Q1 (92.0% favor LLH) to +1.127 dB in Q2 (52.0%), −0.087 dB in Q3 (41.3%), and −0.029 dB in Q4 (45.3%). The FG-LPIPS mean changes are −0.03745, −0.01665, −0.00670, and −0.00520 from Q1 to Q4; the favorable-object rates decline from 97.3% to 50.7%. These strata describe heterogeneity in this cohort. They do not establish that texture complexity causes the response, define a threshold, validate an object selector, or measure texture fidelity.

## S13. Paired color-pathway diagnosis and failure boundaries

The color diagnosis is a selected, failure-enriched set of six earlier cases and 20 Fresh C panels; it is neither random nor a new holdout. No Adapter, GFL, and LLH were compared on the same inputs, checkpoint, and runner across six unique target views (78 object-condition records). CIEDE2000 was recomputed from hash-identified RGB and frozen masks. Intervals are descriptive paired object bootstraps (20,000 resamples, seed 61008); no confirmatory multiplicity family was defined. Figure 6 shows one of the 20 Fresh C panels (UID `f6979d7ff8e54c4daf00fee46f09a139`); its pre-existing C3–GFL stratum was lowest FG-LPIPS, not color-based, and it is distinct from the legacy pink-shift cases.

Table S18: Selected paired endpoints from the failure-enriched 26-object color diagnosis. Each row is condition A minus B with an object-level 95% paired-bootstrap CI. Lower CIEDE2000 and FG-LPIPS are favorable; higher PSNR and Edge-SSIM are favorable. Local texture statistics describe variation, not fidelity.

| Contrast | Endpoint | Mean difference | 95% CI |
|---|---|---|---|
| GFL – No Adapter | Foreground CIEDE2000 | −19.654 | [−26.370, −13.248] |
| GFL – No Adapter | FG-PSNR (dB) | +6.646 | [+5.556, +7.635] |
| GFL – No Adapter | FG-LPIPS | −0.03251 | [−0.05363, −0.01072] |
| GFL – No Adapter | Edge-SSIM | +0.06510 | [+0.04853, +0.08202] |
| LLH – GFL | Foreground CIEDE2000 | +3.583 | [+0.462, +6.712] |
| LLH – GFL | FG-PSNR (dB) | −0.197 | [−1.562, +1.297] |
| LLH – GFL | FG-LPIPS | −0.01294 | [−0.02742, +0.00066] |
| LLH – GFL | Edge-SSIM | −0.00080 | [−0.00776, +0.00659] |
| LLH – GFL | Foreground gradient magnitude | −0.06772 | [−0.08132, −0.05494] |
| LLH – GFL | Foreground high-frequency energy | −0.00887 | [−0.01091, −0.00672] |
| LLH – GFL | Foreground color entropy | −0.598 | [−0.703, −0.502] |

GFL lowered reference-relative CIEDE2000 in 24/26 objects, but visible pink or purple remained in two legacy cases under all three conditions. LLH increased mean color error while reducing local variation, not fidelity. The RGB shift was visible before baking and at denoising update 40. For one case, a ground-truth VAE round trip gave CIEDE2000 4.086 and $\Delta a^* \approx -1.35$, versus 22.032 and +1.977 for GFL; this weakens a VAE-only explanation but does not identify the cause.

A locked P0 trace of three preselected objects reproduced saved RGB pixels exactly in repeated Python 3.13.5 runs; Python 3.10 reproduced its corresponding archived observer. The first cross-runtime tensor difference followed multiplication of the shared FP16 initial latent by the scheduler noise scale (maximum absolute difference 0.0625; RMSE 0.00966; 32.3% changed). This resolves output identity only for these three traces; it establishes neither cohort-wide cross-stack equivalence nor the color cause.

The feature-relative residual gate increased mean CIEDE2000 by 1.35 across 26 objects and 3.18 in the six earlier cases, with no clear visual improvement. The complete cause remains unresolved; no repair is validated.

## S16. Matched applied-budget comparison and revision validity corrections

The same-cohort GeoTex addendum contains 300 objects by six conditions, 1,800 rows. Each object uses the same initial latent, input tensors and checkpoint across conditions. Early-, Middle- and Late-high use 16 high updates at 0–15, 17–32 and 34–49. Low/high are 1.25/2.50, Fixed Mean is 1.65, and the native shallow cap is 0.8. Their exact applied per-group sums are 82.5, 82.5 and 40. This comparison measures the total effect of allocating the high segment at equal applied scalar budget. Evolving residual norms are outcomes of the trajectory; lack of equal residual norms limits a mechanism interpretation, not validity of the allocation contrast. The study is post-inspection on the already evaluated Fresh C objects, not an

independent confirmation.

Table S19: Selected paired contrasts from the 300-object equal-budget addendum. Mean A minus B and pointwise 95% object-bootstrap interval (50,000 resamples). The other 59 pair–metric contrasts form one post hoc Holm family; no interval is multiplicity-adjusted.

| Contrast | Endpoint | Mean [95% CI] | Holm $p$ |
|---|---|---|---|
| Middle − Fixed Applied Mean | fg-psnr | $-0.13326$ $[-0.16979, -0.09622]$ | 1e-05 (raw) |
| Middle − Fixed Applied Mean | fg-lpips | $+0.00199$ $[+0.00147, +0.00254]$ | 0.00059 |
| Middle − Fixed Applied Mean | fg-ssim | $-0.01185$ $[-0.01347, -0.01025]$ | 0.00059 |
| Middle − Fixed Applied Mean | edge-ssim | $-0.00703$ $[-0.00764, -0.00643]$ | 0.00059 |
| Middle − Late-high | fg-psnr | $-0.56395$ $[-0.63408, -0.49427]$ | 0.00059 |
| Middle − Late-high | fg-lpips | $+0.00466$ $[+0.00381, +0.00553]$ | 0.00059 |
| Middle − Late-high | fg-ssim | $-0.00422$ $[-0.00694, -0.00139]$ | 0.01375 |
| Middle − Late-high | edge-ssim | $-0.01409$ $[-0.01526, -0.01291]$ | 0.00059 |
| Middle − Early-high | fg-psnr | $+0.49619$ $[+0.37195, +0.62282]$ | 0.00059 |
| Middle − Early-high | fg-lpips | $-0.00472$ $[-0.00568, -0.00378]$ | 0.00059 |
| Middle − Early-high | fg-ssim | $+0.00499$ $[+0.00211, +0.00809]$ | 0.00432 |
| Middle − Early-high | edge-ssim | $+0.00080$ $[-0.00008, +0.00170]$ | 0.1565 |
| Middle − Fixed High | fg-psnr | $-0.74767$ $[-0.88995, -0.60691]$ | 0.00059 |
| Middle − Fixed High | fg-lpips | $+0.01070$ $[+0.00889, +0.01255]$ | 0.00059 |
| Middle − Fixed High | fg-ssim | $-0.04553$ $[-0.04937, -0.04165]$ | 0.00059 |
| Middle − Fixed High | edge-ssim | $-0.01881$ $[-0.02083, -0.01678]$ | 0.00059 |

The named Middle–Fixed Mean FG-PSNR endpoint is shown with its raw $p$; remaining 59 contrasts use a common post hoc Holm correction in the delivered recomputation. All are retrospective. The rank counts in main matched-budget table are not cross-metric superiority tests.

**Validity corrections to earlier native experiments.** The earlier seven-object MV-Adapter temporal archive passed source-path identity, but used Fresh C targets with a different mesh normalization and orthographic range. In an input-only native geometry audit, one object's six foreground IoUs were about 0.21–0.61. Its old quality estimates are therefore not validated native-camera cross-backbone evidence; the original identity test remains valid. The new targets use the exact native mesh loader, orthographic cameras and nvdiffrast rasterizer, with a source embedded base-color texture and white background. This is unlit albedo, not a claim of PBR or source-render lighting equivalence.

The earlier five-object bake rotated generated pole-view RGB back into raw-camera coordinates but also incorrectly rotated the raw-camera mask. Four of five objects have affected masks. Its old quality estimates are superseded by corrected evaluation and are not pooled with newly selected objects. In the REPEAT branch, bilinear fractions must be calculated before wrapping integer indices: the previous order could produce negative weights at tile edges and outside [0,1]. Corrected code passes nonnegative/unit-sum weight checks and integer-period translation invariance. Scalar coverage masks are read without an RGB transfer function; decoding grayscale coverage as sRGB would change the foreground threshold. GT source images are resized to 256 using bicubic RGB and the same nearest-index alpha masks as generated sources; atlas, camera, fill and render parameters are shared. The original GT-512 bake is retained only as an unmatched-resolution diagnostic.

**Default-path identity boundary.** For GeoTex, explicit requested scale 1.0 and an unset scale are different: the cap is applied only when the scale attribute exists. Executing the actual wrapper with a unit residual gives deep/middle/shallow applied factors (1,1,0.8) for explicit 1.0, versus (1,1,1) when unset. The stored condition name `native_default_1x` therefore denotes Constant 1.0 (capped), not an unmodified pipeline. This is a wrapper-level counterexample, not a claim of full-pipeline pixel equality. In contrast, the new MV-Adapter identity gate compares three textured meshes and 18 images: official default, deployed default and a constant-1 schedule produce bitwise-identical RGB pixels, condition-encoder features and injected residuals. The 13 texture-absent MV-Adapter inputs use the documented geometry-only loader

compatibility branch; their targets are constant white albedo and remain in the primary cohort.

## S17. Frozen native MV-Adapter comparison on fifty new-to-interface objects

The protocol was internally frozen before new scheduled outputs; it is not a public preregistration. Fifty inputs were selected in UID order from Fresh C after excluding every recorded previously inspected MV-Adapter UID. Selection uses available inputs and native geometry/material criteria, not quality or parameter search. Seeds are 20261010 + index, shared across conditions. Six native-camera 512-pixel images use 50 DDPM-shift updates, CFG 3, reference strength 1 and the frozen high-quality text prompt. All 350 rows completed. Models, input identities, runner, native-target calibration and per-step residual traces are indexed in the reproduction archive.

Early-, Middle- and Late-high use high=1 over steps 0–15, 17–32 and 34–49 respectively, with low=0.75 elsewhere. Fixed Mean=0.83 and Generic Linear spans 0.75–0.91 inclusively; each sums to 41.5 per feature, whereas Native/Fixed High sums to 50 and Fixed Low to 37.5. These are feature-scaling controls on MV-Adapter, not GeoTex's per-ResNet capped residual doses.

Official/deployed default and constant-1 scheduled paths match exactly on three textured new objects, including 18 RGB images, encoded conditions and injected residuals. Native target geometry positions/masks match the control rasterization for all fifty inputs; an independent CPU bilinear source-UV sample agrees within the frozen tolerance. Thirteen texture-absent assets use geometry-only compatibility loading and white target albedo; they remain in the primary fifty. The 37-object textured subset is descriptive. It does not create a second primary or substitute after observing outcomes.

The primary Middle-high–Fixed Mean FG-PSNR mean is +0.02899 dB, 95% CI $[-0.01941, +0.08074]$, Monte Carlo sign-flip $p = 0.28075$ (100,000 draws, plus-one correction); 29/50 favor Middle. Intervals use 50,000 object resamples. The other 104 contrasts (21 condition pairs by five endpoints, minus primary) form one predeclared Holm family; intervals remain pointwise. No non-inferiority or architecture-causal claim follows from a non-separated contrast.

Table S20: Complete native fifty-object pair–metric contrasts. A minus B; lower LPIPS/CIEDE2000 and higher other endpoints are favorable. $p$ is Holm-adjusted except the single primary marked raw.

| A – B | Endpoint | Mean [95% CI] | $p$ |
|---|---|---|---|
| High – Low | fg-psnr | −0.12682 [−0.24503, −0.02980] | 0.85601 |
| High – Low | fg-lpips | +0.00100 [−0.00197, +0.00592] | 1.00000 |
| High – Low | fg-ssim | −0.00250 [−0.00594, +0.00029] | 1.00000 |
| High – Low | edge-ssim | +0.00511 [+0.00114, +0.00979] | 0.85601 |
| High – Low | ciede2000 | +0.44730 [+0.13967, +0.87441] | 0.20880 |
| High – Mean | fg-psnr | −0.08309 [−0.16282, −0.02065] | 0.78170 |
| High – Mean | fg-lpips | +0.00013 [−0.00120, +0.00214] | 1.00000 |
| High – Mean | fg-ssim | −0.00268 [−0.00534, −0.00051] | 1.00000 |
| High – Mean | edge-ssim | +0.00163 [−0.00023, +0.00339] | 1.00000 |
| High – Mean | ciede2000 | +0.33298 [+0.10737, +0.65171] | 0.13932 |
| High – Early | fg-psnr | +0.00563 [−0.03952, +0.05131] | 1.00000 |
| High – Early | fg-lpips | −0.00158 [−0.00243, −0.00081] | 0.01649 |
| High – Early | fg-ssim | +0.00002 [−0.00176, +0.00171] | 1.00000 |
| High – Early | edge-ssim | +0.00228 [−0.00040, +0.00477] | 1.00000 |
| High – Early | ciede2000 | +0.03229 [−0.06895, +0.13756] | 1.00000 |
| High – Middle | fg-psnr | −0.11208 [−0.22257, −0.02534] | 0.89608 |
| High – Middle | fg-lpips | +0.00175 [−0.00105, +0.00664] | 1.00000 |
| High – Middle | fg-ssim | −0.00200 [−0.00524, +0.00047] | 1.00000 |
| High – Middle | edge-ssim | +0.00562 [+0.00250, +0.00961] | 0.02112 |
| High – Middle | ciede2000 | +0.41481 [+0.13251, +0.81706] | 0.14875 |
| High – Late | fg-psnr | −0.14046 [−0.25619, −0.04753] | 0.26910 |
| High – Late | fg-lpips | +0.00203 [−0.00108, +0.00734] | 1.00000 |
| High – Late | fg-ssim | −0.00341 [−0.00673, −0.00079] | 0.87002 |
| High – Late | edge-ssim | +0.00119 [−0.00198, +0.00545] | 1.00000 |
| High – Late | ciede2000 | +0.47923 [+0.17811, +0.90198] | 0.05490 |
| High – Linear | fg-psnr | −0.12216 [−0.22718, −0.04049] | 0.22752 |
| High – Linear | fg-lpips | +0.00174 [−0.00098, +0.00639] | 1.00000 |

| A – B | Endpoint | Mean [95% CI] | $p$ |
|---|---|---|---|
| High – Linear | fg-ssim | −0.00309 [−0.00618, −0.00068] | 0.90959 |
| High – Linear | edge-ssim | +0.00243 [−0.00040, +0.00605] | 1.00000 |
| High – Linear | ciede2000 | +0.42004 [+0.14924, +0.81173] | 0.05005 |
| Low – Mean | fg-psnr | +0.04373 [−0.00118, +0.09206] | 1.00000 |
| Low – Mean | fg-lpips | −0.00087 [−0.00381, +0.00085] | 1.00000 |
| Low – Mean | fg-ssim | −0.00017 [−0.00118, +0.00090] | 1.00000 |
| Low – Mean | edge-ssim | −0.00348 [−0.00757, −0.00081] | 0.36480 |
| Low – Mean | ciede2000 | −0.11432 [−0.24502, −0.00126] | 1.00000 |
| Low – Early | fg-psnr | +0.13245 [+0.04720, +0.24287] | 0.15288 |
| Low – Early | fg-lpips | −0.00258 [−0.00777, +0.00036] | 1.00000 |
| Low – Early | fg-ssim | +0.00253 [+0.00006, +0.00594] | 1.00000 |
| Low – Early | edge-ssim | −0.00283 [−0.00657, −0.00016] | 1.00000 |
| Low – Early | ciede2000 | −0.41501 [−0.81018, −0.14622] | 0.04876 |
| Low – Middle | fg-psnr | +0.01475 [−0.01731, +0.04840] | 1.00000 |
| Low – Middle | fg-lpips | +0.00075 [+0.00030, +0.00122] | 0.17384 |
| Low – Middle | fg-ssim | +0.00050 [−0.00056, +0.00160] | 1.00000 |
| Low – Middle | edge-ssim | +0.00051 [−0.00072, +0.00181] | 1.00000 |
| Low – Middle | ciede2000 | −0.03249 [−0.10291, +0.03715] | 1.00000 |
| Low – Late | fg-psnr | −0.01363 [−0.02840, +0.00083] | 1.00000 |
| Low – Late | fg-lpips | +0.00103 [+0.00051, +0.00167] | 0.00800 |
| Low – Late | fg-ssim | −0.00091 [−0.00156, −0.00021] | 0.79142 |
| Low – Late | edge-ssim | −0.00392 [−0.00547, −0.00234] | 0.00303 |
| Low – Late | ciede2000 | +0.03193 [+0.00504, +0.05951] | 1.00000 |
| Low – Linear | fg-psnr | +0.00467 [−0.02786, +0.03373] | 1.00000 |
| Low – Linear | fg-lpips | +0.00074 [+0.00032, +0.00118] | 0.09680 |
| Low – Linear | fg-ssim | −0.00058 [−0.00154, +0.00033] | 1.00000 |
| Low – Linear | edge-ssim | −0.00267 [−0.00418, −0.00118] | 0.10701 |
| Low – Linear | ciede2000 | −0.02726 [−0.09723, +0.04844] | 1.00000 |
| Mean – Early | fg-psnr | +0.08872 [+0.03375, +0.16183] | 0.07565 |
| Mean – Early | fg-lpips | −0.00171 [−0.00411, −0.00028] | 0.42750 |
| Mean – Early | fg-ssim | +0.00270 [+0.00064, +0.00545] | 0.78170 |
| Mean – Early | edge-ssim | +0.00065 [−0.00092, +0.00210] | 1.00000 |
| Mean – Early | ciede2000 | −0.30069 [−0.59160, −0.11032] | 0.02280 |
| Middle – Mean | fg-psnr | +0.02899 [−0.01941, +0.08074] | 0.28075 (raw) |
| Mean – Middle | fg-lpips | +0.00162 [−0.00005, +0.00462] | 1.00000 |
| Mean – Middle | fg-ssim | +0.00067 [−0.00064, +0.00204] | 1.00000 |
| Mean – Middle | edge-ssim | +0.00399 [+0.00170, +0.00764] | 0.00104 |
| Mean – Middle | ciede2000 | +0.08184 [−0.02823, +0.20315] | 1.00000 |
| Mean – Late | fg-psnr | −0.05737 [−0.10476, −0.01422] | 0.85601 |
| Mean – Late | fg-lpips | +0.00190 [−0.00003, +0.00530] | 1.00000 |
| Mean – Late | fg-ssim | −0.00074 [−0.00183, +0.00031] | 1.00000 |
| Mean – Late | edge-ssim | −0.00044 [−0.00301, +0.00363] | 1.00000 |
| Mean – Late | ciede2000 | +0.14625 [+0.03581, +0.27574] | 0.87002 |
| Mean – Linear | fg-psnr | −0.03907 [−0.07392, −0.01081] | 0.57941 |
| Mean – Linear | fg-lpips | +0.00161 [+0.00009, +0.00434] | 1.00000 |
| Mean – Linear | fg-ssim | −0.00041 [−0.00129, +0.00045] | 1.00000 |
| Mean – Linear | edge-ssim | +0.00081 [−0.00123, +0.00410] | 1.00000 |
| Mean – Linear | ciede2000 | +0.08707 [+0.01651, +0.17259] | 1.00000 |
| Early – Middle | fg-psnr | −0.11771 [−0.23017, −0.02649] | 0.98234 |
| Early – Middle | fg-lpips | +0.00333 [+0.00036, +0.00863] | 0.57941 |
| Early – Middle | fg-ssim | −0.00203 [−0.00568, +0.00081] | 1.00000 |
| Early – Middle | edge-ssim | +0.00334 [+0.00057, +0.00690] | 1.00000 |
| Early – Middle | ciede2000 | +0.38253 [+0.11622, +0.76613] | 0.19845 |
| Early – Late | fg-psnr | −0.14609 [−0.25687, −0.06083] | 0.03572 |
| Early – Late | fg-lpips | +0.00361 [+0.00036, +0.00933] | 0.80715 |
| Early – Late | fg-ssim | −0.00344 [−0.00694, −0.00080] | 0.80715 |
| Early – Late | edge-ssim | −0.00109 [−0.00429, +0.00294] | 1.00000 |
| Early – Late | ciede2000 | +0.44694 [+0.17823, +0.84174] | 0.01089 |
| Early – Linear | fg-psnr | −0.12778 [−0.22984, −0.04995] | 0.03572 |
| Early – Linear | fg-lpips | +0.00332 [+0.00045, +0.00838] | 0.31416 |
| Early – Linear | fg-ssim | −0.00311 [−0.00648, −0.00061] | 1.00000 |
| Early – Linear | edge-ssim | +0.00016 [−0.00259, +0.00353] | 1.00000 |
| Early – Linear | ciede2000 | +0.38776 [+0.14561, +0.75366] | 0.01176 |
| Middle – Late | fg-psnr | −0.02838 [−0.05857, −0.00004] | 1.00000 |

| A – B | Endpoint | Mean [95% CI] | $p$ |
|---|---|---|---|
| Middle – Late | fg-lpips | +0.00028 [−0.00018, +0.00080] | 1.00000 |
| Middle – Late | fg-ssim | −0.00141 [−0.00243, −0.00042] | 0.66671 |
| Middle – Late | edge-ssim | −0.00443 [−0.00568, −0.00316] | 0.00104 |
| Middle – Late | ciede2000 | +0.06442 [−0.00064, +0.12764] | 1.00000 |
| Middle – Linear | fg-psnr | −0.01008 [−0.04413, +0.02065] | 1.00000 |
| Middle – Linear | fg-lpips | −0.00001 [−0.00041, +0.00033] | 1.00000 |
| Middle – Linear | fg-ssim | −0.00109 [−0.00220, −0.00019] | 1.00000 |
| Middle – Linear | edge-ssim | −0.00318 [−0.00412, −0.00227] | 0.00104 |
| Middle – Linear | ciede2000 | +0.00523 [−0.07199, +0.08600] | 1.00000 |
| Late – Linear | fg-psnr | +0.01830 [−0.01050, +0.04259] | 1.00000 |
| Late – Linear | fg-lpips | −0.00029 [−0.00101, +0.00020] | 1.00000 |
| Late – Linear | fg-ssim | +0.00033 [−0.00046, +0.00100] | 1.00000 |
| Late – Linear | edge-ssim | +0.00125 [+0.00038, +0.00194] | 0.15288 |
| Late – Linear | ciede2000 | −0.05919 [−0.12493, +0.00991] | 1.00000 |

Table S21: Descriptive 37-object textured subset, retained separately from the primary fifty. No subset inference is promoted to confirmation.

| Condition | PSNR | LPIPS | SSIM | Edge-SSIM | CIEDE2000 |
|---|---|---|---|---|---|
| High | 14.06 | 0.1159 | 0.4354 | 0.3239 | 18.04 |
| Low | 14.14 | 0.1170 | 0.4345 | 0.3174 | 17.82 |
| Mean | 14.11 | 0.1165 | 0.4356 | 0.3204 | 17.87 |
| Early | 14.06 | 0.1168 | 0.4345 | 0.3182 | 17.99 |
| Middle | 14.12 | 0.1165 | 0.4347 | 0.3182 | 17.84 |
| Late | 14.16 | 0.1164 | 0.4359 | 0.3233 | 17.79 |
| Linear | 14.14 | 0.1164 | 0.4360 | 0.3217 | 17.82 |

## S18. Calibrated bake pilot, historical separation and actual-output displays

The new bake list contains ten objects, chosen in frozen UID order before observing their bake results. All 300 Fresh C inputs were screened; old five-object outputs and the previously inspected REPEAT diagnostic object were excluded. Eligibility required available source GLB, one opaque white-factor base-color material, native TEXCOORD_0, no vertex colors, at most 300,000 triangles, and source camera-mask alignment passing the frozen gate. This input-selected set is smaller than the target twenty and below fifteen; we retain the pilot designation and all ten outcomes. No failed or adverse-quality object was replaced. Seven conditions generated 70 object-condition rows; eight bake conditions including GT generated 80 artifacts, with 880 unseen renders and 480 source-camera renders.

The frozen primary was Late-high–Fixed Mean FG-LPIPS. C3–GFH PSNR, LPIPS, SSIM and CIEDE2000 form one four-test secondary Holm family. All other complete pairs are descriptive. Views are averaged before inference. Intervals use 50,000 paired object resamples and are pointwise, not simultaneous. The ten-object primary uses all $2^{10}$ sign flips; the secondary family uses the same exact test. A sign-flip test assumes symmetry/exchangeability of paired differences under its null; an interval alone is not a practical-equivalence criterion. No non-inferiority margin was chosen.

Table S22: Prespecified new-pilot comparisons: raw A minus B, 95% object-bootstrap interval; no pooling with old five.

| Contrast | Endpoint | Mean [95% CI] | $p$ |
|---|---|---|---|
| Late–Mean | FG-LPIPS | $-0.00265$ $[-0.00414, -0.00137]$ | 0.00195 (primary) |
| C3–GFH | FG-PSNR | $+0.193$ $[-0.649, +0.875]$ | 1.000 (Holm) |
| C3–GFH | FG-LPIPS | $-0.00036$ $[-0.00396, +0.00336]$ | 1.000 (Holm) |
| C3–GFH | FG-SSIM | $-0.05274$ $[-0.07459, -0.03170]$ | 0.00781 (Holm) |
| C3–GFH | CIEDE2000 | $-1.102$ $[-2.915, +0.939]$ | 0.95508 (Holm) |

The corrected earlier five-object bake uses the same source-resolution and mask corrections but retains its historical clamped-UV technical scope. Its descriptive means (PSNR / LPIPS / SSIM) are GFH 11.86 / 0.1609 / 0.5513, C3 14.24 / 0.1542 / 0.4170, Fixed Mean 14.19 / 0.1538 / 0.4348 and Late-high 15.02 / 0.1479 / 0.4428. C3–GFH is +2.37 dB, not the superseded +2.27 dB estimate. There was no independent predeclared primary for this reevaluation. The older N=20 result, old five, new ten and corrected single-object REPEAT diagnostic each retain separate source identifiers. Earlier wrong-mask, gamma-decoded-alpha and wrong-REPEAT-weight artifacts remain audit records and are superseded, never combined with the valid reevaluation.

Table S23: Per-object generated-view versus unseen-bake rankings across seven conditions. Bootstrap unit is object. Positive overall agreement does not require an identical winning condition.

| Endpoint | Mean Spearman $\rho$ [95% CI] | Winner changed |
|---|---|---|
| FG-PSNR | 0.846 [0.696, 0.968] | 1/10 |
| FG-LPIPS | 0.804 [0.693, 0.911] | 7/10 |
| FG-SSIM | 0.746 [0.450, 0.964] | 4/10 |
| Edge-SSIM | 0.907 [0.871, 0.943] | 6/10 |

GT and generated bake sources use the same six raw cameras and 256-pixel resolution. GT RGB uses bicubic downsampling; alpha uses the exact nearest-index mapping used for generated foreground inputs. Their six-view scalar masks are bitwise identical and their raw atlas coverage matches for every object and condition. No unseen view enters baking. Export checks retain binary geometry/UV buffers, transforms and mesh identity. REPEAT UV weights are nonnegative, sum to one and are invariant under integer UV shifts. These checks establish a valid unlit projection comparison, not PBR equivalence or a validated

seam score. Texture boundaries can be natural appearance boundaries; seam diagnostics are not reported as a validated primary endpoint.

Supplementary Video presents two input-order, rights-cleared objects at all eleven discrete unseen cameras. Each actual rendered view is held for one second; no interpolated camera image is synthesized. It is an unlit comparison, not a claim of continuous-view or PBR quality.

LG Remote by mohd.akmal.mahyudin; CC Attribution. Source: https://sketchfab.com/3d-models/1dc2c857f7614caeb6c2aa0559ccf973.

Tire by Koto3D Stephane Vogley; CC Attribution. Source: https://sketchfab.com/3d-models/21678bd623654182b52c11d013de4e71.

Asset 238ff3e41c964655b83aa9c8770cc766 by suntaree; CC Attribution. Source: https://sketchfab.com/3d-models/238ff3e41c964655b83aa9c8770cc766.

Baptisimal Font - Aughrim by Galway3D_DH_AGe; CC0 Public Domain. Source: https://sketchfab.com/3d-models/2dd9bc8172c6461797947cc7214db399.

Pittsburgh Bridge Collapse in Frick Park by C8X; CC Attribution. Source: https://sketchfab.com/3d-models/2f6c94487ee54cb88021f4643409d849.

bricks on news boxes by bunnyhero; CC Attribution. Source: https://sketchfab.com/3d-models/3c83796ca5f043f1beb8044dd2e51a0d.

Eastern Box Turtle Shell by Frank McMains; CC Attribution. Source: https://sketchfab.com/3d-models/85a7df80b79a4ba9b8c3a175de0dbce6.

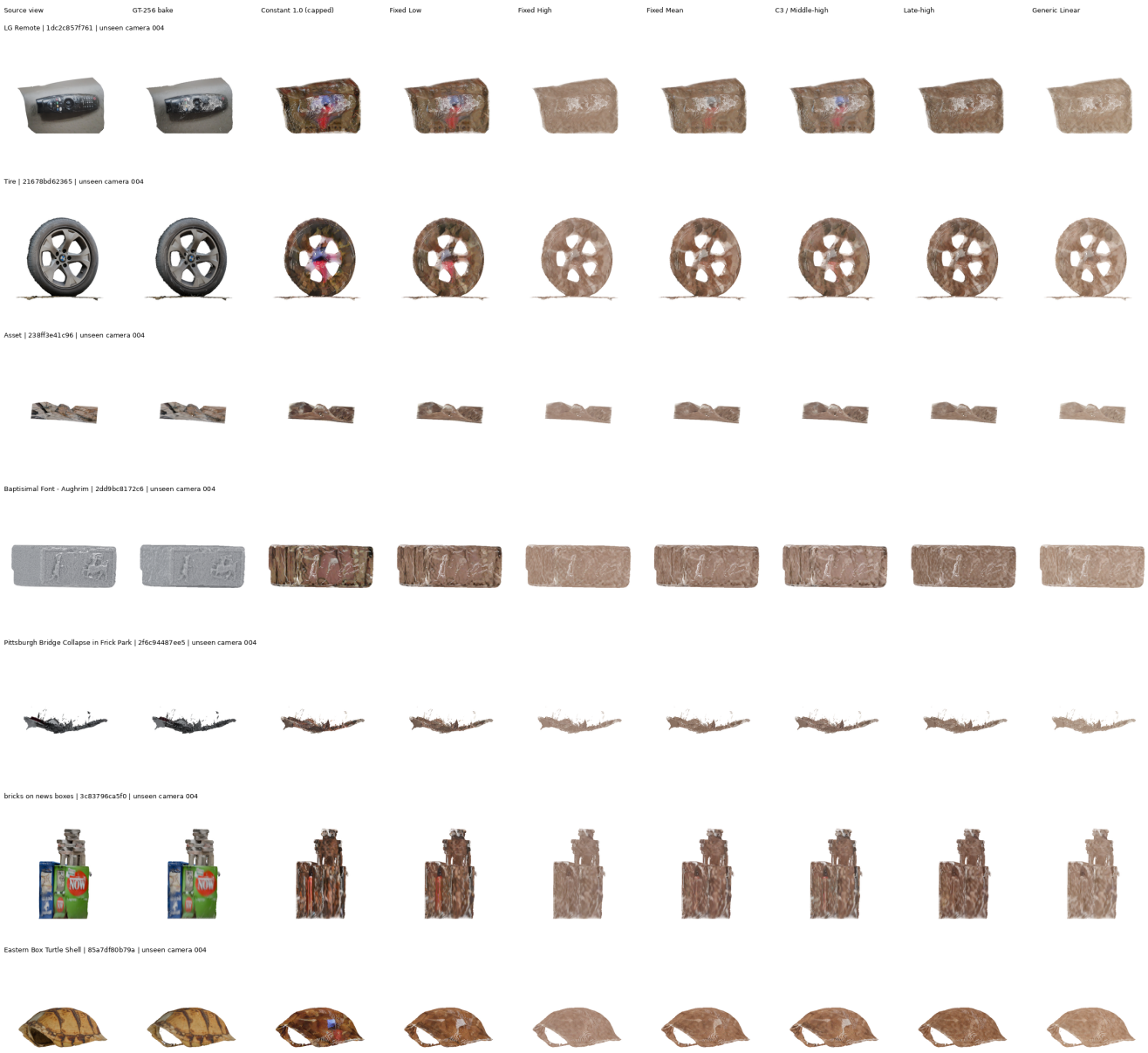



Figure S2: Full-object same-camera source, GT-256 bake and all seven controls for the seven new-pilot objects whose publication licenses were verified. Three other objects remain in all numerical analyses but are omitted from distributed illustrations because image-publication permission is unresolved. No quality-based exclusion is made. Images are actual renders, resized for display without retouching. Source credits and input/output hashes are provided in BAKED_ASSET_CREDITS.csv and BAKED_FIGURE_VIDEO_PROVENANCE.csv.

## S19. Post hoc GT reference context and object influence

This reanalysis uses the same valid ten-object source records and the same frozen endpoints; it is not a new holdout or a new confirmatory family. GT-256 is a matched empirical projection reference, not a theoretical upper bound on an image endpoint. Its mean PSNR/LPIPS/SSIM/Edge-SSIM/CIEDE2000 are 22.38/0.0590/0.7683/0.7533/4.44. Late-high's LPIPS gain relative to Fixed Mean is unchanged at 0.00265.

For descriptive magnitude context, the ratio of means is

$$r = \frac{\text{mean}(\text{LPIPS}_{Mean} - \text{LPIPS}_{Late})}{\text{mean}(\text{LPIPS}_{Mean} - \text{LPIPS}_{GT256})} = 0.03720.$$

The numerator and denominator are recomputed jointly in each of 50,000 object-bootstrap samples (seed 20261010). All bootstrap denominators are positive; the pointwise interval is [0.02275,0.05323]. This is a roughly 3.7% reduction of the mean empirical reference gap, not evidence that the remaining appearance mismatch is small. It must not be interpreted as a practical acceptability threshold or a new primary discovery. Removing each object in turn yields Late–Mean mean differences between −0.00294 and −0.00208; this descriptive influence check does not create additional independent samples.

Table S24: Same ten objects: matched reference and generated unseen-view FG-LPIPS. All eleven unseen views are averaged first. Full UIDs and exact values accompany the reanalysis script.

| UID prefix | GT-256 | Fixed Mean | Late-high | Late–Mean |
|---|---|---|---|---|
| 1dc2c857f761 | 0.07674 | 0.15307 | 0.14898 | -0.00408 |
| 21678bd62365 | 0.07367 | 0.17198 | 0.16919 | -0.00279 |
| 238ff3e41c96 | 0.02328 | 0.05895 | 0.05849 | -0.00046 |
| 2dd9bc8172c6 | 0.14110 | 0.23763 | 0.22982 | -0.00781 |
| 2f6c94487ee5 | 0.06525 | 0.09319 | 0.09317 | -0.00003 |
| 3c83796ca5f0 | 0.05597 | 0.17757 | 0.17439 | -0.00318 |
| 85a7df80b79a | 0.04051 | 0.10964 | 0.10519 | -0.00445 |
| e261c0fe85af | 0.04032 | 0.13280 | 0.13112 | -0.00168 |
| f1ccc55a65c7 | 0.06034 | 0.11759 | 0.11622 | -0.00137 |
| fa5c4b02c5b2 | 0.01312 | 0.05075 | 0.05010 | -0.00065 |

The script `scripts/analyze_gt_context.py`, copied source CSV and `GT_REFERENCE_CONTEXT.json` permit standalone reproduction with NumPy. No generated output is added, replaced or retouched. The source hash identifies the previous round's valid metrics; historical five, N=20 and invalid old native-seven records remain separate.